7.3 Minimisation Methods

Candidate number: *insert candidate number*

insert date of final run

Computational Setup

In 3–5 sentences, state the language/environment, double precision, how λ^* was selected or checked, and how iteration data were recorded. Mention any convergence checks only if they affected the reported runs. Avoid implementation detail unless it explains a numerical result.

Table 1: Summary of reported runs.

Run	Function	Method	Starting point	Iterations reported
Q2	(4)	Steepest descents	$(\cdots, \cdots)$	$\cdots$
Q3	(5)	Steepest descents	$(\cdots, \cdots)$	$\cdots$
Q4	(4)	Conjugate gradients	$(\cdots, \cdots)$	$\cdots$
Q5	(5)	Conjugate gradients	$(\cdots, \cdots)$	$\cdots$
Q6	(6)	DFP	$(\cdots, \cdots, \cdots)$	$\cdots$
Q7	(4)	DFP	$(\cdots, \cdots)$	$\cdots$
Q8	(5)	DFP	$(\cdots, \cdots)$	$\cdots$

Question 1: Shape and Analytic Minima

Table 2: Analytic minima of the two two-variable objectives.

Function	Minimiser (x^*, y^*)	f^*
(4)	$(-0.3700, -0.7937)$	-1.0953
(5)	$(1, 1)$	0

Function (4). Solving $\partial_x f = 1 + x - y^2 = 0$ for $x = y^2 - 1$ and substituting into $\partial_y f = 1 - 2y + 4y(y^2 - x/2) = 0$ gives $1 + 2y^3 = 0$. Since $y \mapsto 2y^3$ is strictly increasing on $\mathbb{R}$, the equation has the unique real root $y^* = -2^{-1/3}$, so $x^* = 2^{-2/3} - 1$. The Hessian at (x^*, y^*) is $H = \left(\begin{smallmatrix} 1 & 2^{2/3} \\ 2^{2/3} & 10 \cdot 2^{-2/3} \end{smallmatrix}\right)$ with $\det H = 6 \cdot 2^{-2/3} = 3 \cdot 2^{1/3} \approx 3.78 > 0$ and $\operatorname{tr} H > 0$; by the second-derivative test, (x^*, y^*) is a local minimum. Coercivity: in polar coordinates the quartic $(y^2 - x/2)^2$ has leading term $r^4 \sin^4 \theta$, which dominates for $\sin\theta \neq 0$; along $\sin\theta = 0$ (i.e. $y = 0$), $f = x^2/4 + x \to \infty$ as $|x| \to \infty$. Hence the unique stationary point is the global minimum. Eliminating $x = y^2 - 1$ reduces f to $g(y) = -\frac{1}{2} + y + y^4/2$, giving $f^* = g(y^*) = -\frac{1}{2} - 2^{-1/3} + 2^{-7/3} \approx -1.0953$.

Function (5). $\forall x, y \in \mathbb{R}$, $f = (1 - x)^2 + 80(y - x^2)^2 \geq 0$ with equality iff $y = x^2$ and $x = 1$. This corresponds to $(x^*, y^*) = (1, 1)$ and $f^* = 0$, making it a unique global minimum.

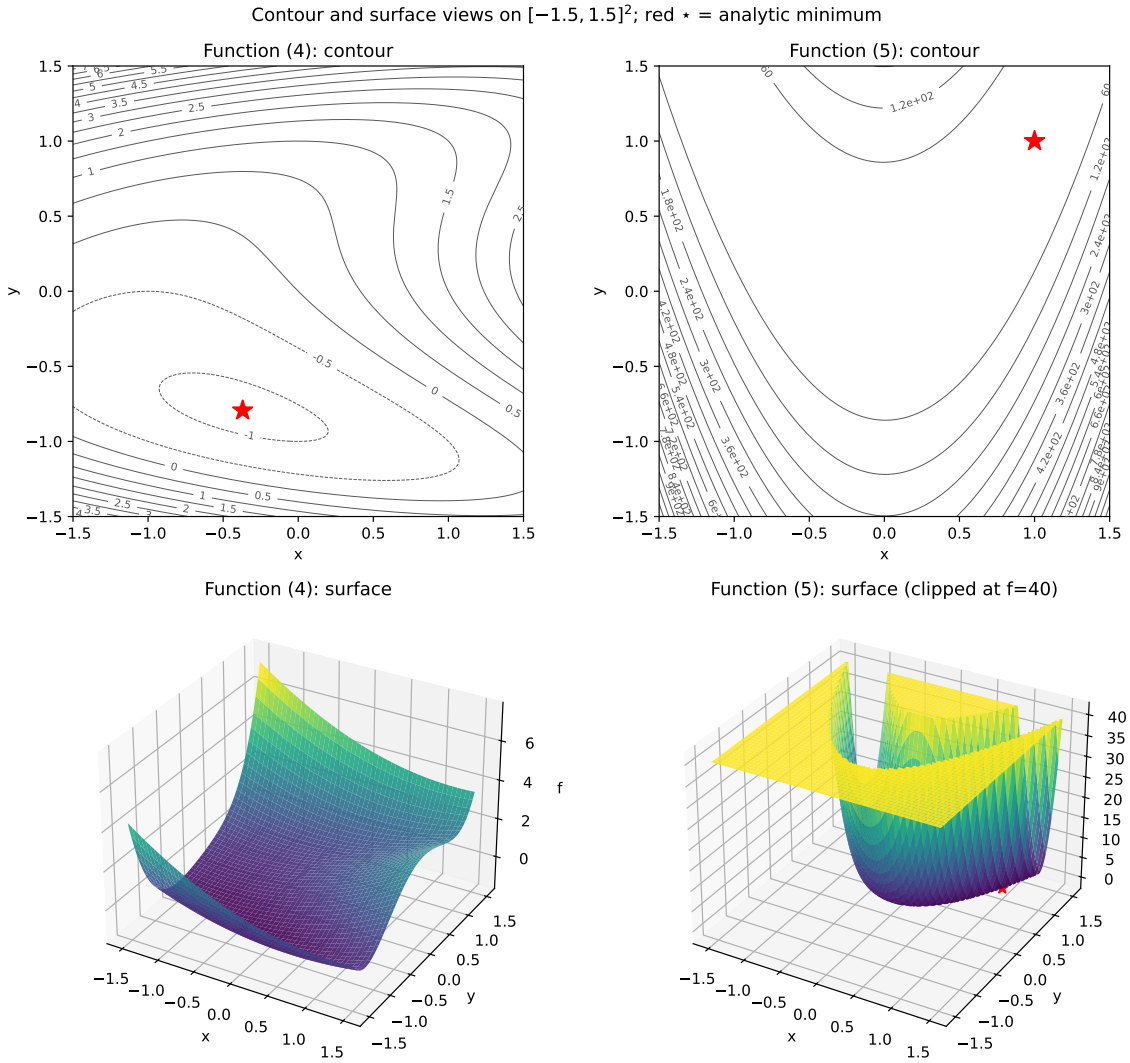

Figure 1: Contour and surface plots of functions (4) and (5) on $[-1.5, 1.5]^2$; red stars mark the analytic minima.

Questions 2–3: Steepest Descents

Question 2: Function (4)

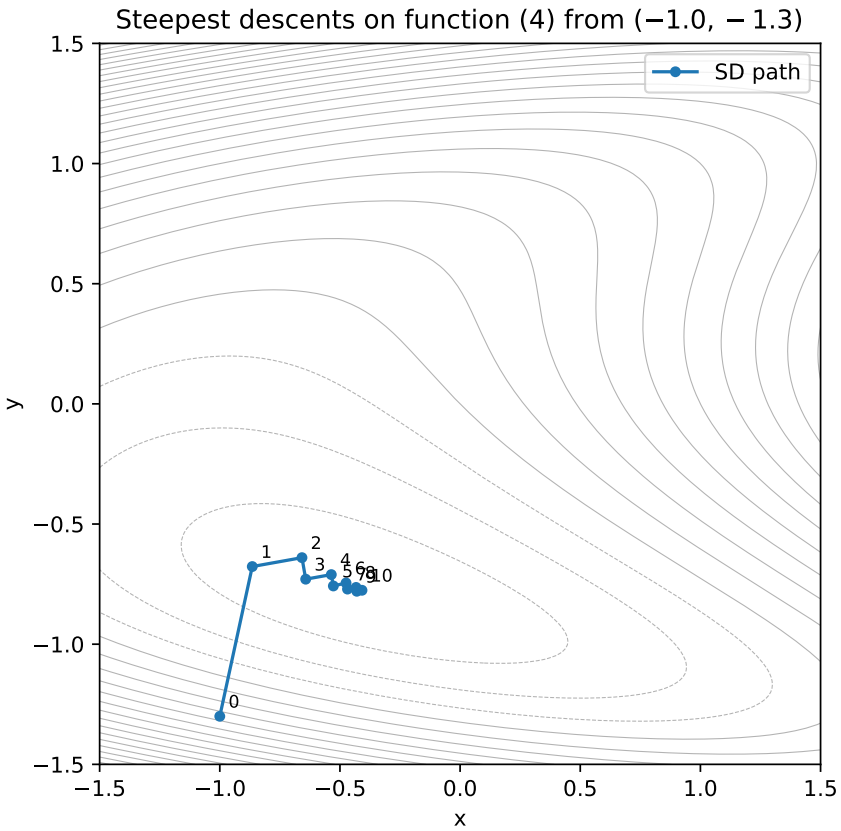

Figure 2: Steepest-descents path on function (4) from $(-1.0, -1.3)$; ten iterations, numbered, with contours of f for reference.

Line-search quantisation. The 201-point grid on $\lambda \in [0, 2]$ with $\Delta\lambda = 10^{-2}$ gives a per-step error bounded by $\frac{1}{8}\phi''(\lambda^*)(\Delta\lambda)^2 \lesssim 10^{-4}$, well below the observed $|f_n - f^*| \sim 10^{-3}$. Quantisation is therefore not the limiting error and is ignored hereafter.

Linear tail and value estimate. On a convex quadratic, by the Kantorovich inequality [3]:

$$f_{n+1} - f^* \;\le\; \left(\frac{\kappa - 1}{\kappa + 1}\right)^2 (f_n - f^*), \qquad \kappa = \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)},$$

Hence consider $r_n = |\Delta f_{n+1}|/|\Delta f_n|$ from Table 3:

n	2	3	4	5	6	7	8	9
r_n	0.554	0.509	0.575	0.513	0.625	0.655	0.609	0.602

giving $r \approx 0.60$. Inverting the Kantorovich bound supplies a lower bound $\kappa \gtrsim 8$ on the local condition number; the true value at x^* (computed from the Hessian in Q1: $\lambda_{\min} \approx 0.56$, $\lambda_{\max} \approx 6.74$) is $\kappa \approx 12$, so the observed rate is below but compatible with the worst-case bound. Modelling the tail as geometric, $f_n - f^* \approx Cr^n \approx (r/(1-r))|\Delta f_n|$; substituting $|\Delta f_{10}| = 4.24 \times 10^{-4}$ and $r \in [0.58, 0.66]$ gives $f_{10} - f^* \in [5.9, 7.9] \times 10^{-4}$. Hence

$$\boxed{f^* \approx -1.0953, \qquad f^* \in [-1.096, -1.094].}$$

Table 3: Steepest-descents trace for function (4) from $(-1.0, -1.3)$. Line search: grid search on $\lambda \in [0, 2]$ with step 0.01, giving λ^* to 2 s.f.

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n
0	-1.0000	-1.3000	–	$+1.0561$	–
1	-0.8648	-0.6770	0.08	-1.0198	-2.076
2	-0.6580	-0.6399	0.64	-1.0538	-3.40×10^{-2}
3	-0.6425	-0.7295	0.23	-1.0726	-1.89×10^{-2}
4	-0.5360	-0.7104	0.61	-1.0822	-9.59×10^{-3}
5	-0.5274	-0.7577	0.21	-1.0878	-5.51×10^{-3}
6	-0.4746	-0.7453	0.52	-1.0906	-2.83×10^{-3}
7	-0.4686	-0.7707	0.20	-1.0923	-1.77×10^{-3}
8	-0.4329	-0.7638	0.57	-1.0935	-1.16×10^{-3}
9	-0.4298	-0.7797	0.19	-1.0942	-7.04×10^{-4}
10	-0.4083	-0.7757	0.57	-1.0946	-4.24×10^{-4}

Coordinate estimates. *(i)* x_n monotone increasing, bounded above by x^*. Even-step differences $\Delta_2 x_n = x_n - x_{n-2}$: 0.207, 0.107, 0.053, 0.035, 0.022. Average even-step decay rate (denoted as ρ_x) ≈ 0.58. Model $x_n = x^* - D\rho_x^{n/2}$; single-step tail from $n = 11$ is two interleaved GPs of common ratio ρ_x per two iterations. Since $\Delta x_{11} \approx \rho_x \cdot \Delta x_9$ and $\Delta x_{12} \approx \rho_x \cdot \Delta x_{10}$:

$$x^* - x_{10} = \frac{\rho_x \cdot \Delta x_9}{1 - \rho_x} + \frac{\rho_x \cdot \Delta x_{10}}{1 - \rho_x} = \frac{\rho_x}{1 - \rho_x}(\Delta x_{10} + \Delta x_9) \approx 0.034,$$

so $x^* \approx -0.37$, and from Range $\rho_x \in [0.50, 0.75]$: $x^* \in [-0.45, -0.30]$.

(ii) Odd and even subsequences of y_n *both* monotone decreasing; between-parity gaps halve: 0.037, 0.020, 0.013, 0.007, 0.004. Hence both bound y^* from above; $y^* \leq y_{10} = -0.7757$. Odd-subseq differences 0.053, 0.028, 0.013, 0.009, ratio $\rho_y \approx 0.56$. Extrapolate from y_9:

$$y^* \approx y_9 - \frac{\rho_y}{1 - \rho_y} \cdot 0.009 = -0.7797 - \frac{0.56}{0.44} \cdot 0.009 \approx -0.792, \qquad y^* \in [-0.82, -0.76].$$

Value precision vs point precision. After 10 iterations: f^* to 4 s.f. ($\varepsilon_f \sim 6 \times 10^{-4}$); (x^*, y^*) to 1–2 s.f. ($|\varepsilon| \sim 4 \times 10^{-2}$). The observation $\varepsilon_f \sim |\varepsilon|^2$ up to $O(1)$ shows one digit of value precision is gained from a half-digit (square-root) of point precision.

Why. Since $g(x^*) = 0$, Taylor expansion at $x = x^* + \varepsilon$ gives $f(x^* + \varepsilon) = f^* + \frac{1}{2}\varepsilon^T H \varepsilon + O(|\varepsilon|^3)$: the linear term vanishes. Applying the Rayleigh-quotient bounds $\lambda_{\min}(H)|\varepsilon|^2 \leq \varepsilon^T H \varepsilon \leq \lambda_{\max}(H)|\varepsilon|^2$ then yields

$$\sqrt{\frac{2\,\varepsilon_f}{\lambda_{\max}(H)}} \lesssim |\varepsilon| \lesssim \sqrt{\frac{2\,\varepsilon_f}{\lambda_{\min}(H)}}\,.$$

Substituting $\varepsilon_f \sim 6 \times 10^{-4}$ and the Q1 eigenvalues $\lambda_{\min} \approx 0.56$, $\lambda_{\max} \approx 6.74$ bounds $|\varepsilon| \in [0.013, 0.046]$, consistent with the observed $|\varepsilon| \sim 4 \times 10^{-2}$. The relation $\varepsilon_f \sim |\varepsilon|^2$ is universal at smooth non-degenerate minima: f is flat to first order and curved to second, so location precision is the binding constraint at fixed iteration budget — not line-search quantisation, not value conditioning.

Question 3: Function (5)

Line-search quantisation. Finer grid is required because $\lambda_1^* \approx 2 \times 10^{-3}$ lies below Q2's $\Delta\lambda = 10^{-2}$: the start is just below the valley ($0.676^2 = 0.4570 > 0.443$), so g_0 has a large

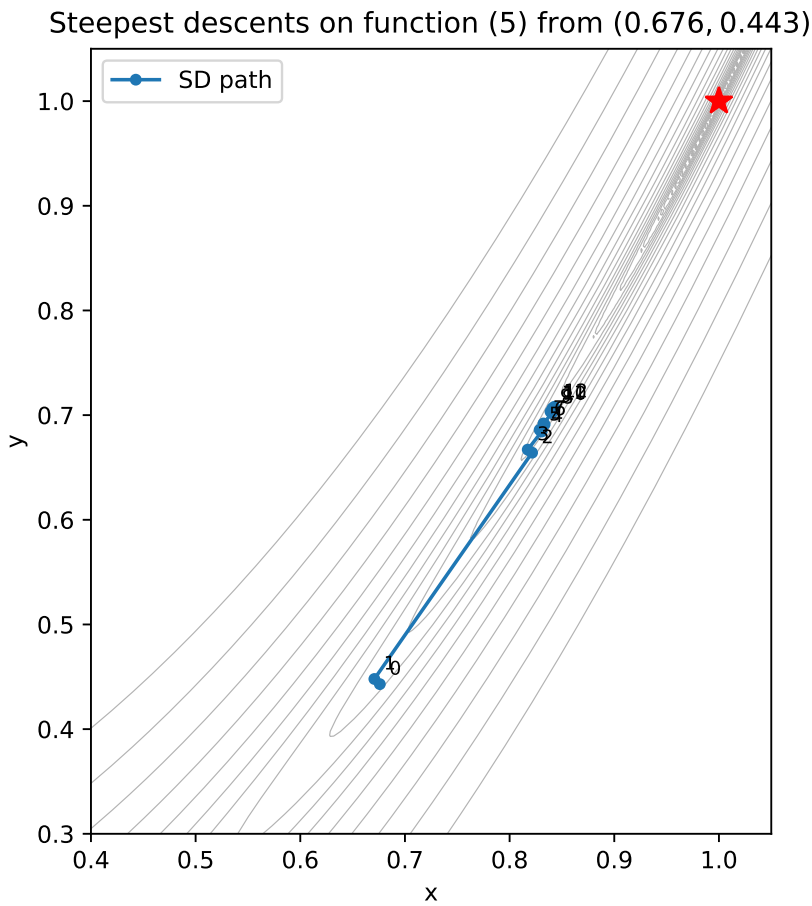

Figure 3: Steepest-descents path on function (5) from $(0.676, 0.443)$; twelve iterations, numbered, with contours of f (log-spaced). Red $\star$ marks $(1, 1)$. After step 1 the iterates cluster inside $[0.82, 0.85] \times [0.66, 0.71]$, crawling along the curved valley $y = x^2$.

Table 4: Steepest-descents trace for function (5) from $(0.676, 0.443)$. Line search: grid on $\lambda \in [0, 2]$ with $\Delta\lambda = 10^{-4}$ (20001 points), finer than Q2's 10^{-2}.

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n
0	+0.6760	+0.4430	–	+0.12060	–
1	+0.6708	+0.4479	0.0022	+0.10872	-1.19×10^{-2}
2	+0.8215	+0.6641	0.6694	+0.04104	-6.77×10^{-2}
3	+0.8173	+0.6670	0.0017	+0.03345	-7.59×10^{-3}
4	+0.8307	+0.6844	0.1141	+0.03128	-2.17×10^{-3}
5	+0.8287	+0.6859	0.0017	+0.02940	-1.88×10^{-3}
6	+0.8334	+0.6910	0.0386	+0.02876	-6.35×10^{-4}
7	+0.8324	+0.6920	0.0017	+0.02816	-6.09×10^{-4}
8	+0.8405	+0.7021	0.0744	+0.02701	-1.15×10^{-3}
9	+0.8391	+0.7033	0.0017	+0.02595	-1.06×10^{-3}
10	+0.8416	+0.7057	0.0206	+0.02564	-3.04×10^{-4}
11	+0.8409	+0.7065	0.0018	+0.02534	-2.98×10^{-4}
12	+0.8427	+0.7080	0.0136	+0.02514	-2.04×10^{-4}

y-component and the search line crosses the steep valley wall quickly. With $\Delta\lambda = 10^{-4}$ the error in λ^* is $\leq 5 \times 10^{-5}$; per-step error in f is bounded by $\frac{1}{8}\phi''(\lambda^*)(\Delta\lambda)^2 \lesssim 10^{-9}$, far below observed $|\Delta f| \geq 10^{-4}$. Quantisation is not the limiting error.

Limit point. The unique stationary point of f is $(1,1)$: $\partial_y f = 160(y - x^2) = 0 \Rightarrow y = x^2$, then $\partial_x f = -2(1-x) = 0 \Rightarrow x = 1$. The level set $\{f \leq f_0\}$ is bounded, since $f \geq (1-x)^2$ gives $|x| \leq 1 + \sqrt{f_0}$ and $f \geq 80(y - x^2)^2$ then bounds $|y|$. Hence (x_n) is bounded; f_n is monotone decreasing and bounded below by 0, so $\Delta f_n \to 0$, and the standard SD bound $\Delta f_n \geq c|g_n|^2$ (for $c > 0$ depending on local Hessian bounds) gives $|g_n| \to 0$. Every subsequential limit is therefore stationary, and uniqueness forces $x_n \to (1,1)$, $f^* = 0$.

Linear tail and rate. λ_n^* alternates: odd-n steps have $\lambda_n^* \approx 0.002$ (iterate sits near $y = x^2$, $g_y \approx 0$, the short step along $-g_x$ is constrained by the curving floor); even-n steps have λ_n^* larger as the iterate drifts off-valley and $-g$ points back across. This is the zigzag across the narrow curved valley. Subsampling at even n to remove zigzag noise:

n	4	6	8	10	12
f_n/f_{n-2}	0.763	0.920	0.938	0.948	0.981

f_n/f_{n-2} is *increasing* toward the two-step Kantorovich bound [3] $r_K^2 = ((\kappa-1)/(\kappa+1))^4 \approx 1 - 8/\kappa$: at $(1,1)$, $H = \left(\begin{smallmatrix} 642 & -320 \\ -320 & 160 \end{smallmatrix}\right)$ has eigenvalues ≈ 801.6, 0.399, $\kappa \approx 2.0 \times 10^3$, $r_K \approx 0.998$; at the actual iterate $(x_{12}, y_{12}) = (0.843, 0.708)$, direct calculation gives $\kappa \approx 0.85 \times 10^3$, $r_K \approx 0.995$ (the small eigenvalue depends approximately linearly on $y - x^2$, so κ varies between $\sim 10^2$ and $\sim 4{\times}10^3$ across the cluster). Using the late per-iteration rate $r \approx 0.99$ in $f_n - f^* \approx |\Delta f_n|/(1-r)$:

$$f_{12} - f^* \approx 2 \times 10^{-2}, \qquad f^* \approx f_{12} - 2 \times 10^{-2} \approx 5 \times 10^{-3}.$$

This central estimate is uncertain because κ (hence r) varies along the curved valley, so the fixed-r geometric model is only a local surrogate. Combined with the analytic lower bound $f^* \geq 0$:

$$\boxed{f^* \in [0, \sim 10^{-2}]\,.}$$

Coordinate precision via the valley identity. Iterates lie approximately on $y = x^2$, so $f_n = (1-x_n)^2 + 80(y_n - x_n^2)^2$ is dominated by the first term: at $n = 12$, $(1-0.843)^2 = 0.0246 \approx f_{12} = 0.0251$ (the 2% residual is the small $(y - x^2)$ offset). Hence $|1 - x_n| \approx \sqrt{f_n}$ gives a direct location estimate tighter than any interleaved-GP extrapolation: $|1 - x_{12}| \approx 0.158$, so $x^* \in [0.84, 1]$ and by the valley $y^* \in [0.71, 1]$; 12 iterations have covered only $(1 - 0.158/0.324) \approx 51\%$ of the along-valley distance. The precision relation $\varepsilon_f \sim |\varepsilon|^2$ holds here in its *along-valley* form with constant 1 (not $\frac{1}{2}\lambda_{\max}$), because $f(x, x^2) = (1-x)^2$ has second derivative 2.

Sensitivity to λ^*.

Per-step $\pm 5\%$ perturbation gives final points: baseline $(0.843, 0.708)$, $f = 0.025$; $+5\% \to (0.688, 0.470)$, $f = 0.098$; $-5\% \to (0.879, 0.772)$, $f = 0.015$. The $+5\%$ run overshoots at the breakthrough step and climbs the opposite valley wall; the -5% run undershoots and stays on the near wall. Linearisation at exact line-search minimum: since $\phi'(\lambda^*) = 0$, a perturbation $\delta\lambda$ gives $\phi(\lambda^* + \delta\lambda) - \phi(\lambda^*) = \frac{1}{2}\phi''(\lambda^*)\delta\lambda^2 + O(\delta\lambda^3)$, so the per-step *$f$-value* error is $O(\delta\lambda^2)$ on any smooth f. The per-step *position* error, however, is $O(\delta\lambda)$: $\delta x_{n+1} = \delta\lambda_n s_n$, hence $\delta g_{n+1} \approx -\delta\lambda_n H g_n$, growing by $\|H\| \cdot \delta\lambda$. On a tight curved valley like (5), this $O(\delta\lambda)$ position drift is not absorbed because the next baseline step depends sensitively on the iterate's distance from the valley curve $y = x^2$; a $\delta\lambda$ kick translates into a different segment of the long valley, and over many steps the cumulative drift dominates. Line-search accuracy is therefore binding for path stability, though not for per-step value precision.

Why SD is inefficient. Three reinforcing causes, each visible in Figure 3: *(a) Anisotropy.* Along-valley curvature is $d^2/dx^2(1-x)^2 = 2$, cross-valley is $\partial^2 f/\partial y^2 = 160$, giving an axis-aligned curvature ratio ≈ 80. The Hessian's principal axes at $(1,1)$ are not aligned with (x, y),

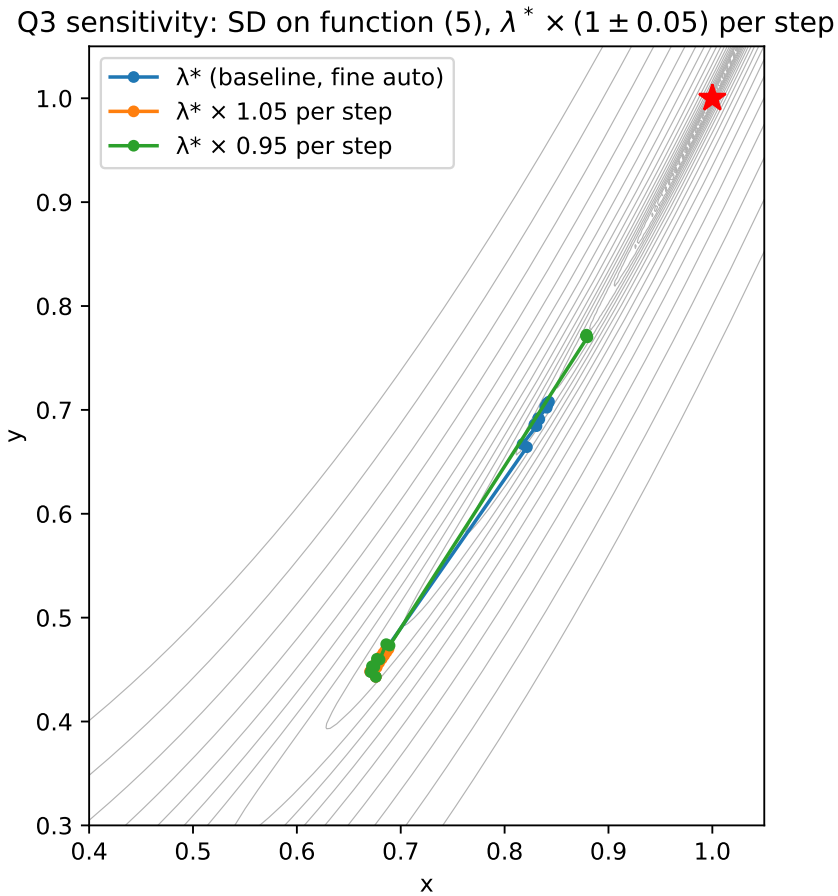

Figure 4: SD on (5): baseline path and $\pm 5\%$ per-step perturbations of λ^*. Paths diverge from $n = 2$, where the baseline step $\lambda_2^* = 0.67$ amplifies any multiplicative perturbation directly into $|\Delta x| \sim 0.03$.

so the spectral condition number is larger: $\kappa \approx 2 \times 10^3$, $r_K \approx 1 - 4/\kappa \approx 0.998$ asymptotically. *(b) Wrong direction.* Off-valley, $\partial f/\partial y = 160(y - x^2)$ dominates $\partial f/\partial x$, so $-g$ points across the valley, not along it — SD picks the wrong direction. *(c) Short steps.* $\lambda^* \approx |g|^2/(g^T H g)$ with $g^T H g \sim 160\, g_y^2$ large forces λ^* small; along-valley progress per iteration is $O(1/\kappa)$ of cross-valley motion, producing the zigzag. Conjugate gradients (Q5) break (b) by search-direction memory; DFP (Q8) breaks (a,c) by absorbing curvature into a variable metric.

Questions 4–5: Conjugate Gradients

Question 4: Function (4)

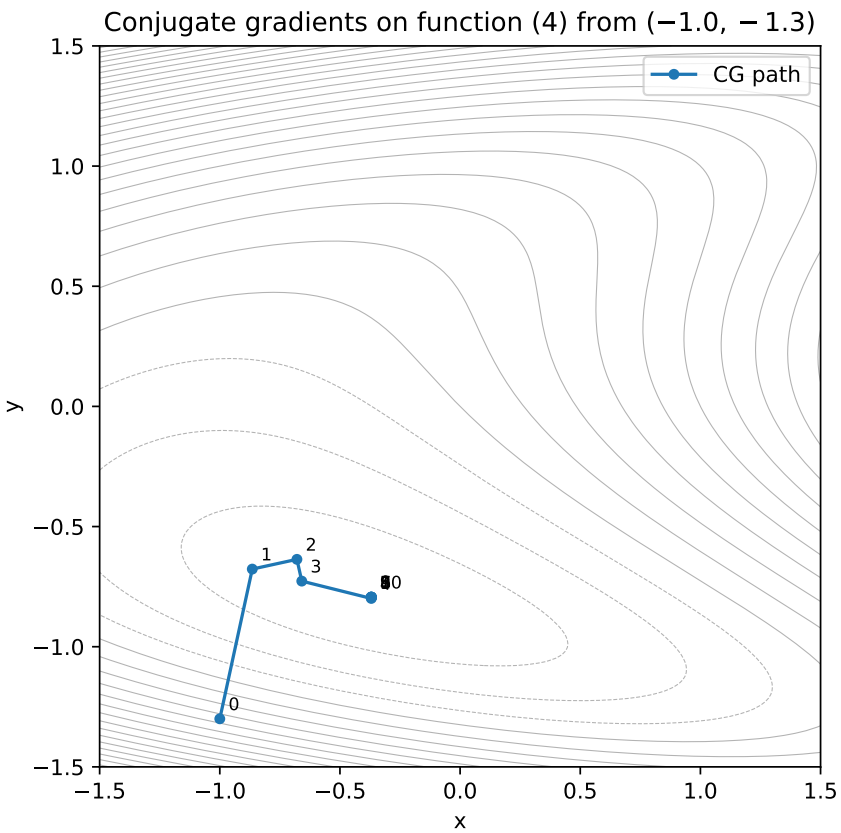

Figure 5: Conjugate-gradient path on function (4) from $(-1.0, -1.3)$; ten iterations, same start and line-search grid as Q2.

Finite termination on quadratics. For $f(x) = \frac{1}{2}(x - x^*)^T A\,(x - x^*) + c$ with $A \succ 0$, exact line search at step k gives $g_k^T s_{k-1} = 0$; combined with the recursion (7) and induction on k this yields mutual gradient orthogonality $g_k^T s_j = 0$ for $j < k$ and direction A-conjugacy $s_i^T A s_j = 0$ for $i \neq j$ (cf. [1]). A-orthogonal vectors are linearly independent, so $\{s_0, \ldots, s_{N-1}\}$ spans $\mathbb{R}^N$ and $g_N = 0$. Here $N = 2$ and the code restarts every N steps: each pair (odd SD-restart, even CG) therefore *exactly* minimises any local quadratic model in two iterations, the key to the behaviour below.

Three regimes in Table 5.

- *Pre-quadratic* ($n = 1$–3): the iterate is ~ 0.4 from x^* and the quartic term $(y^2 - x/2)^2$ dominates, so the local Hessian model is inaccurate. CG beats SD only marginally: compare $\Delta f_2 = -3.1 \times 10^{-2}$ with Q2's -3.4×10^{-2}.

Table 5: Conjugate-gradient trace for function (4) from $(-1.0, -1.3)$. Same line search as Q2 (grid on $\lambda \in [0, 2]$ with spacing 0.01). Restart every $N = 2$ steps: odd-n steps are SD (restarts), even-n steps use the conjugate recurrence (7).

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n	step
0	-1.0000	-1.3000	–	$+1.0561$	–	–
1	-0.8648	-0.6770	0.08	-1.0198	-2.076	SD
2	-0.6790	-0.6364	0.57	-1.0509	-3.11×10^{-2}	CG
3	-0.6588	-0.7270	0.24	-1.0698	-1.89×10^{-2}	SD
4	-0.3696	-0.7991	1.39	-1.0952	-2.54×10^{-2}	CG
5	-0.3684	-0.7941	0.15	-1.09527	-8.89×10^{-5}	SD
6	-0.3688	-0.7942	0.43	-1.09527	-2.82×10^{-7}	CG
7	-0.3689	-0.7940	0.20	-1.09528	-1.81×10^{-7}	SD
8	-0.3700	-0.7937	1.32	-1.09528	-3.79×10^{-7}	CG
9	-0.3700	-0.7937	0.15	-1.09528	-4.10×10^{-10}	SD
10	-0.3700	-0.7937	1.10	-1.09528	-4.59×10^{-12}	CG

- *Break-out* ($n = 4$, CG): a single step of length $\|x_4 - x_3\| \approx 0.30$ with $\lambda_4^* = 1.39$ moves $(-0.659, -0.727) \mapsto (-0.370, -0.799)$, essentially onto x^*. This is finite termination at work on the quadratic model fitted at $n = 3, 4$.
- *Quadratic clean-up* ($n = 5$–10): with $|\varepsilon| \lesssim 10^{-3}$ the cubic Taylor remainder is $O(|\varepsilon|^3) \lesssim 10^{-9}$, so each (SD, CG) pair picks up only that residual. This produces the bursts $|\Delta f_6| \sim 10^{-7}$, $|\Delta f_{10}| \sim 10^{-12}$; per-pair contraction of $|\Delta f|$ is $\sim 10^{-5}$ — super-linear, not geometric.

Line-search quantisation. On the Q2 grid ($\Delta\lambda = 10^{-2}$) the per-step bound $\frac{1}{8}\phi''(\lambda^*)(\Delta\lambda)^2 \lesssim 10^{-4}$ is unchanged but exceeds $|\Delta f_n|$ for $n \geq 5$ by many orders. CG still converges because $\phi(\lambda)$ is so flat near x^* that the value is insensitive to λ^* (e.g. $\lambda_6^* = 0.43$ yields $|\Delta f_6| = 2.8 \times 10^{-7}$); the finite-termination cascade is robust to the 2 s.f. line search.

Numerical estimates from the trace alone. Imagining the analytic answer is unknown: five-digit agreement of f from $n = 6$ onwards and four-digit agreement of (x_n, y_n) from $n = 8$ onwards, together with $|g_{10}| = 4.4 \times 10^{-6}$ and the Q2 Taylor bound $\varepsilon_f \leq \frac{1}{2}\lambda_{\max}(H)|\varepsilon|^2$ (with $\lambda_{\max}(H) \approx 6.3$), give

$$\boxed{f^* = -1.09528 \pm 1 \times 10^{-5}, \qquad x^* \in [-0.3701, -0.3699], \qquad y^* \in [-0.7938, -0.7936].}$$

Compared with the Q2 intervals $f^* \in [-1.096, -1.094]$, $x^* \in [-0.45, -0.30]$, $y^* \in [-0.82, -0.76]$ at the same iteration budget, CG tightens the value by $\sim 10^2$ and each coordinate by $\sim 10^3$. Q2's general statement — that $\varepsilon_f \sim \frac{1}{2}\lambda_{\max}(H)|\varepsilon|^2$, so f^* can be pinned to roughly twice as many digits as (x^*, y^*) because $f \in C^2$ with $H \succ 0$ at x^* — still holds, but with both sides near the numerical floor the squared-precision bonus is now saturated rather than active.

Comparison with Q2.

Interpretation. At Q2's Kantorovich rate $r \approx 0.60$ the linear-tail extrapolation gives $f_{10} - f^* \sim 10^{-3}$, the SD ceiling. CG instead executes a finite-termination pair on each local quadratic model and pays only the $O(|\varepsilon|^3)$ cubic remainder; once x enters the basin where the Hessian model is valid (here at $n = 4$) the linear rate becomes super-linear and the remaining iterations are clean-up. The improvement on function (4) is therefore qualitative, not a constant-factor speed-up.

Table 6: Same start, same 10 iterations, same line search: SD (Q2) vs CG (Q4) on (4).

Quantity at $n = 10$	Q2 (SD)	Q4 (CG)	Improvement factor
f_n	-1.0946	-1.09528	—
$f_n - f^*$	$\sim 6 \times 10^{-4}$	$\lesssim 10^{-8}$	$\sim 10^5$
(x_n, y_n)	$(-0.408, -0.776)$	$(-0.3700, -0.7937)$	—
$\|(x_n, y_n) - (x^*, y^*)\|$	$\sim 4 \times 10^{-2}$	$\sim 10^{-4}$	$\sim 4 \times 10^2$
$\|g_n\|$	5×10^{-2}	4×10^{-6}	$\sim 10^4$
$\|\Delta f_n\|$	4.2×10^{-4}	4.6×10^{-12}	$\sim 10^8$

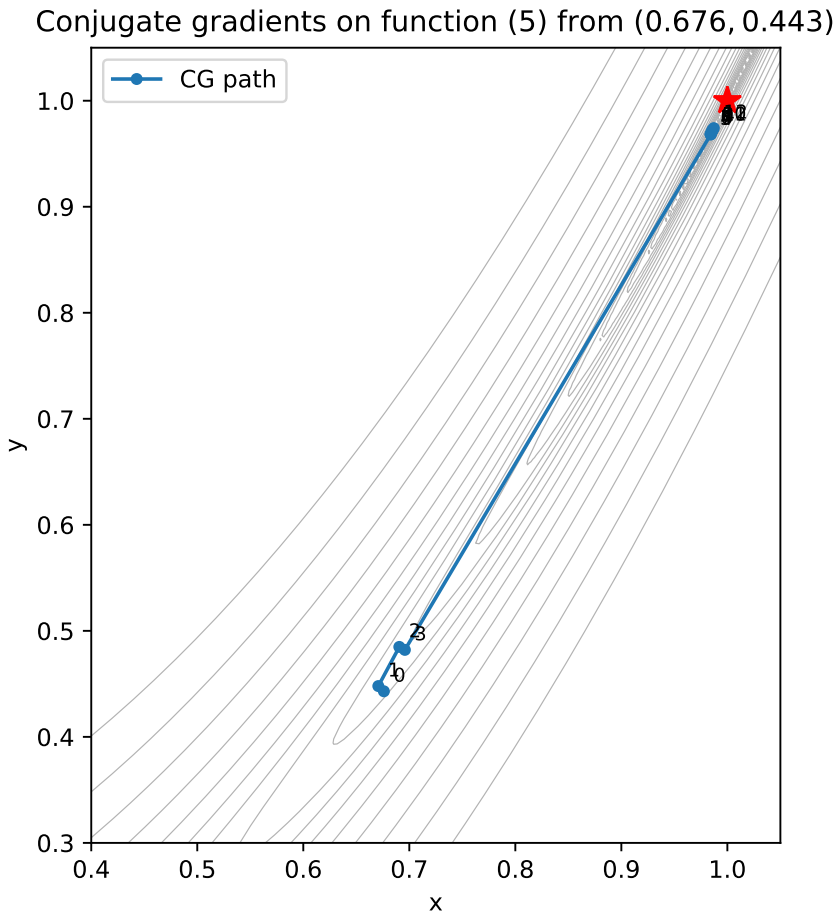

Figure 6: Conjugate-gradient path on function (5) from $(0.676, 0.443)$; twelve iterations, numbered, with log-spaced contours of f on $[0.4, 1.05] \times [0.3, 1.05]$. Red $\star$ marks the analytic minimum $(1, 1)$.

Table 7: Conjugate-gradient trace for function (5) from $(0.676, 0.443)$. Line search: grid on $\lambda \in [0, 2]$ with $\Delta\lambda = 10^{-4}$ (the coarse Q2 grid would miss the $\lambda^* \sim 10^{-3}$ SD steps). With $N = 2$ restart, odd n are SD (including the first step and every N-step restart), even n are CG.

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n	Step
0	+0.6760	+0.4430	–	$+1.2060\times10^{-1}$	–	start
1	+0.6708	+0.4479	0.0022	$+1.0872\times10^{-1}$	-1.189×10^{-2}	SD
2	+0.6906	+0.4849	0.1039	$+1.0083\times10^{-1}$	-7.885×10^{-3}	CG
3	+0.6958	+0.4821	0.0022	$+9.2883\times10^{-2}$	-7.948×10^{-3}	SD
4	+0.9843	+0.9682	1.558	$+2.8472\times10^{-4}$	-9.260×10^{-2}	CG
5	+0.9841	+0.9683	0.0013	$+2.5426\times10^{-4}$	-3.046×10^{-5}	SD
6	+0.9845	+0.9688	0.0443	$+2.4943\times10^{-4}$	-4.827×10^{-6}	CG
7	+0.9844	+0.9689	0.0013	$+2.4481\times10^{-4}$	-4.623×10^{-6}	SD
8	+0.9857	+0.9720	0.2274	$+2.2160\times10^{-4}$	-2.321×10^{-5}	CG
9	+0.9859	+0.9719	0.0013	$+1.9998\times10^{-4}$	-2.161×10^{-5}	SD
10	+0.9860	+0.9723	0.0365	$+1.9683\times10^{-4}$	-3.150×10^{-6}	CG
11	+0.9861	+0.9723	0.0013	$+1.9382\times10^{-4}$	-3.009×10^{-6}	SD
12	+0.9872	+0.9742	0.1673	$+1.8019\times10^{-4}$	-1.363×10^{-5}	CG

Question 5: Function (5)

Limiting point. The line search makes f_n monotone non-increasing and bounded below by 0, so $f_n \downarrow L \geq 0$. f is coercive, so the iterates stay in a compact sub-level set; on $\mathbb{R}^2$, $\nabla f = 0$ forces $y = x^2$ and $x = 1$, so $(1, 1)$ is the unique stationary point. From $n = 4$ onwards the trace places the iterate inside its basin ($\|(x_n, y_n) - (1, 1)\| < 0.04$), and standard descent theory then gives $L = 0$ and $(x_n, y_n) \to (1, 1)$. The per-step coordinates are not strictly monotone at the displayed precision (e.g. $x_5 < x_4,\ y_9 < y_8$), but f_n is, which is what the convergence argument uses.

Finite-termination heuristic and pair-wise rate. On a 2-D quadratic with exact line searches CG reaches the minimum at $n = 2$. Restart every $N = 2$ steps preserves this idea locally: each (SD, CG) pair *is* a full CG cycle on the quadratic model of f near the current point. The breakthrough pair $n = 3 \to 4$ ($\lambda_4^* = 1.558$, close to the bracket limit) shows this in action: the direction $s_4 = -g_3 + \beta\, s_3$ comes out essentially tangent to the valley axis, so a single long step takes (x, y) from $(0.696, 0.482)$ to $(0.984, 0.968)$, collapsing f by a factor ~ 330. The SD direction $-g_3$ alone would cross the valley nearly perpendicularly; the CG correction βs_3 is what rotates it along the valley.

Beyond the breakthrough, f departs from its quadratic model (Rosenbrock's Hessian varies strongly along the valley), so each (SD, CG) pair decrements $e_n = f_n - f^* = f_n$ by a near-constant factor rather than terminating. Reading the post-breakthrough pair residuals e_{2m} for $m = 2, \ldots, 6$ directly from Table 7, the four ratios e_{2m+2}/e_{2m} are 0.876, 0.889, 0.888, 0.915, mean $\rho \approx 0.89$. The geometric extrapolation $f^* \approx f_{12} - \rho/(1-\rho)\,|e_{12} - e_{10}|$ gives $f^* \lesssim 4\times10^{-5}$, consistent with the true $f^* = 0$ and showing the linear tail is the binding constraint at this iteration budget.

Quantitative comparison with Q3. Same start, same 12 iterations: SD (Q3) reaches $f_{12} = 2.51\times10^{-2}$ at $(0.843, 0.708)$; CG reaches $f_{12} = 1.80\times10^{-4}$ at $(0.987, 0.974)$. CG is $\sim 140\times$ smaller in f and $\sim 10\times$ closer in Euclidean distance to $(1, 1)$ (0.32 for SD vs. 0.029 for CG). Almost the entire gap is contributed by the single step $n = 4$; excise it and the remaining progress on $n \in \{1, 2, 3\} \cup \{5, \ldots, 12\}$ is comparable between the two methods.

Asymptotic-rate remark. At $(1, 1)$, Q3's analysis gives $\kappa \approx 2\times10^3$ and $r_{\mathrm{SD}} \approx 0.998$; a

pure-SD tail would need $\sim 10^3$ iterations per decimal of f, consistent with Q3's near-stall. CG with restart terminates in two iterations exactly on the underlying quadratic; here inexact line search and non-quadratic curvature replace this by the finite $\rho \approx 0.89$ seen above, still vastly faster than $r_{\mathrm{SD}}^2 \approx 0.996$ per pair.

Sensitivity to λ^*.

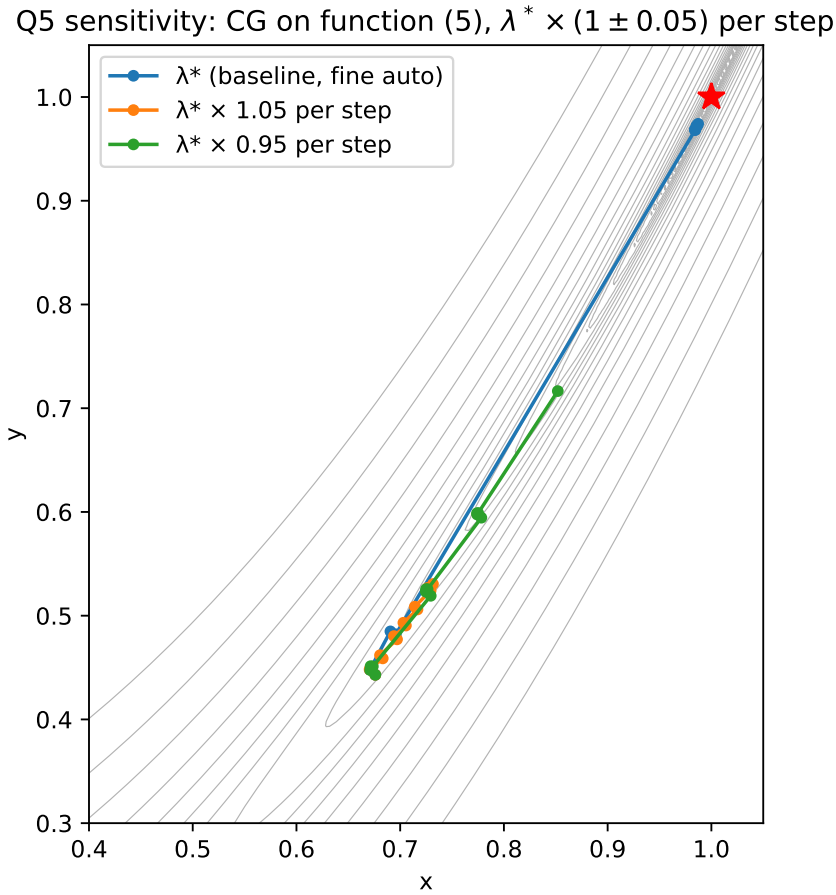

Figure 7: Per-step perturbation: at each step λ^* is recomputed by the fine grid and scaled by 1 ± 0.05. Baseline and perturbed paths share x_1 and already diverge at $n = 2$.

Table 8: CG endpoints after 12 iterations under per-step $\lambda^* \to (1+\delta)\lambda^*$.

δ	Final (x_{12}, y_{12})	$f(x_{12})$	First divergence
0	(0.9872, 0.9742)	1.80×10^{-4}	–
+0.05	(0.7316, 0.5305)	7.38×10^{-2}	$n = 2$
−0.05	(0.8519, 0.7165)	2.88×10^{-2}	$n = 2$

CG is *more* sensitive to λ^* than SD on this function. The CG direction

$$s_k = -g_k + \beta_k\, s_{k-1}, \qquad \beta_k = \frac{g_k \cdot g_k}{g_{k-1} \cdot g_{k-1}},$$

carries both a directional memory s_{k-1} and a scalar memory β_k, each evaluated at iterates already displaced by prior line-search errors. A perturbation δ in λ^*_{k-1} perturbs x_{k-1}, hence g_{k-1}, hence β_k *and* s_k, and is then re-scaled by the new line search. The breakthrough step $n = 4$ is the amplifier: $\lambda^* = 1.558$ traverses a region where the curvature varies by orders of

magnitude, so scaling it by 1.05 overshoots the valley entirely and the $\delta = +0.05$ run ends at $(0.73, 0.53)$ — further from $(1, 1)$ than the SD endpoint of Q3, and even further than the CG starting point. SD has no directional memory, so a mis-scaled step produces a slightly wrong gradient but no cascade; Q3's SD run therefore tolerates the same 5% noise with much smaller endpoint shift.

Does CG offer much improvement over SD? Yes, substantially under accurate line search and conditionally otherwise. On (4) CG pins (x^*, y^*) to four significant figures in ten iterations where SD reaches one (Table 6); on (5) CG cuts f by three orders of magnitude in twelve iterations where SD manages one, the (SD, CG) pair finite-terminating the local quadratic model — the same point Newton's method would reach — even at $\kappa \gtrsim 10^3$. The caveat is brittleness: inexact line searches on strongly non-quadratic functions destroy CG's conjugacy, and a 5% λ^*-error at the breakthrough alone can leave the CG run worse than SD (Table 8); the N-step restart bounds cross-pair damage but cannot repair a broken conjugacy within the current pair.

Questions 6–8: DFP Algorithm

Question 6: Quadratic Function (6)

DFP applied to the quadratic $f(x, y, z) = 0.4x^2 + 0.2y^2 + z^2 + xz$ with constant Hessian $A = \left(\begin{smallmatrix} 0.8 & 0 & 1 \\ 0 & 0.4 & 0 \\ 1 & 0 & 2 \end{smallmatrix}\right)$ (eigenvalues $\{0.234, 0.4, 2.566\}$, $\kappa \approx 10.98$), starting from $x_0 = (1, 1, 1)$ with $H_0 = I$ and the supplied line-search values $\lambda^* = (0.3942, 2.5522, 4.2202)$.

Table 9: DFP trace for the three-dimensional quadratic from $(1, 1, 1)$ with $H_0 = I$.

n	x_n	y_n	z_n	λ_n^*	$f(x_n)$	$\lvert g_n \rvert$
0	+1.000000	+1.000000	+1.000000	–	+2.6000	3.52
1	+0.290440	+0.842320	−0.182600	0.3942	$+1.5595\times10^{-1}$	0.349
2	+0.120098	−0.019152	−0.067979	2.5522	$+2.2998\times10^{-3}$	0.0332
3	$+4.0\times10^{-6}$	-4.2×10^{-5}	$+2.7\times10^{-5}$	4.2202	$+1.188\times10^{-9}$	6.7×10^{-5}

Termination on the quadratic. By Fletcher–Powell [1], on a positive-definite quadratic with exact line search DFP terminates in N steps with $H_N = A^{-1}$ and $g_N = 0$; the secant identity $H^*p = q$ (direct from eqn 8, since $p^T Hp$ is scalar) is the building block. With $N = 3$ this predicts $x^* = 0$, $f^* = 0$ at step 3. Table 9 gives $|x_3| \le 4.2\times10^{-5}$ and $f_3 = 1.19\times10^{-9}$, consistent with the 4-d.p. λ^* truncation: $\delta\lambda \sim 5\times10^{-5}$ propagates (by the quadratic argument below) to $f_3 \sim \frac{1}{2}(\delta\lambda)^2(s_3^T A s_3) \sim 10^{-9}$.

Convergence of H to the inverse Hessian. At step 3 the DFP matrix is

$$H_3 \approx \begin{pmatrix} +3.33333332 & +1.90\times10^{-7} & -1.66666679 \\ +1.90\times10^{-7} & +2.49999784 & +1.37\times10^{-6} \\ -1.66666679 & +1.37\times10^{-6} & +1.33333246 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} \frac{10}{3} & 0 & -\frac{5}{3} \\ 0 & \frac{5}{2} & 0 \\ -\frac{5}{3} & 0 & \frac{4}{3} \end{pmatrix}.$$

Each diagonal agrees with A^{-1} to eight decimals, the six nominally zero off-diagonals are all $O(10^{-6})$, and $\|H_3 - A^{-1}\|_F = 3.04\times10^{-6}$ — consistent with the line-search residual rather than systematic bias.

Invariance to λ_N. The two λ_3 rows of Table 10 show $\|H_3 - A^{-1}\|_F$ unchanged from baseline to all reported digits: this is exact, not numerical. With λ_1, λ_2 at their exact values, H_2 and g_2 are unchanged, so $s_3 = -H_2 g_2$ is unchanged; perturbing $\lambda_3 \to \alpha\lambda_3$ gives $q_3 = \alpha\lambda_3 s_3$ and,

Table 10: Sensitivity of the three-iteration DFP result to per-step perturbation of the supplied λ^* values. Baseline $\lambda^* = (0.3942,\ 2.5522,\ 4.2202)$ gives $x_3 \approx 0$ and $f_3 = 1.19\times10^{-9}$.

Perturbation	x_3	$f(x_3)$	$\|H_3 - A^{-1}\|_F$
none	$(+0.0000,\ -0.0000,\ +0.0000)$	1.19×10^{-9}	3.04×10^{-6}
$\lambda_1\times1.05$	$(-0.0798,\ +0.0802,\ -0.2499)$	8.62×10^{-2}	3.21
$\lambda_1\times0.95$	$(+0.1824,\ +0.1385,\ +0.1870)$	8.62×10^{-2}	3.22
$\lambda_2\times1.05$	$(+0.0274,\ +0.1387,\ -0.0184)$	3.98×10^{-3}	2.88
$\lambda_2\times0.95$	$(-0.0274,\ -0.1388,\ +0.0185)$	3.99×10^{-3}	2.50
$\lambda_3\times1.05$	$(-0.0060,\ +0.0009,\ +0.0034)$	5.75×10^{-6}	3.04×10^{-6}
$\lambda_3\times0.95$	$(+0.0060,\ -0.0010,\ -0.0034)$	5.75×10^{-6}	3.04×10^{-6}
all $\times1.01$	$(-0.0288,\ +0.0208,\ -0.0627)$	6.16×10^{-3}	2.60

because A is constant,

$$p_3 = g(x_2 + \alpha\lambda_3 s_3) - g(x_2) = A\,(\alpha\lambda_3 s_3) = \alpha\, p_3^{\text{base}}.$$

Both correction terms in eqn 8 are then degree-2 in numerator *and* denominator,

$$\frac{q_3 q_3^T}{p_3^T q_3} \propto \frac{\alpha^2}{\alpha^2} = 1, \qquad \frac{H_2 p_3 p_3^T H_2}{p_3^T H_2 p_3} \propto \frac{\alpha^2}{\alpha^2} = 1,$$

so H_3 is *exactly* invariant under $\lambda_3 \to \alpha\lambda_3$ on a quadratic. The *position* is not: $x_3 = x_2 + \alpha\lambda_3 s_3$ is pushed off the minimum, and since the exact λ_3^* minimises $\phi(\lambda) = f(x_2 + \lambda s_3)$ (so $\phi'(\lambda_3^*) = 0$), Taylor about λ_3^* gives

$$f_3(\alpha) \approx \tfrac{1}{2}\,(\alpha - 1)^2\,(\lambda_3^*)^2\,(s_3^T A s_3).$$

Using only Table 9, the exact-line-search identity $\Delta f_3 = -\frac{1}{2}\lambda_3^*\, g_2^T H_2 g_2$ combined with $s_3^T A s_3 = g_2^T H_2 g_2/\lambda_3^*$ gives $s_3^T A s_3 = 2.58\times10^{-4}$; hence at $\alpha = 1.05$,

$$f_3 \approx \tfrac{1}{2}\,(0.05)^2(4.2202)^2(2.58\times10^{-4}) = 5.75\times10^{-6},$$

matching the table to three figures. The leading term is even in $(\alpha - 1)$, so $\pm5\%$ give identical f_3 and (anti-)symmetric x_3, also confirmed.

Sensitivity to λ_1, λ_2. The Fletcher–Powell induction rests on the exact-line-search orthogonality $g_{k+1}^T s_k = 0$, from which A-conjugacy of successive q_k follows. A 5% error in λ_1 violates $g_1^T s_0 = 0$, so q_0 and q_1 cease to be A-conjugate; the relation $H_N p_j = q_j$ fails for $j < N - 1$, and H_3 departs from A^{-1} by $\|H_3 - A^{-1}\|_F \sim 3$, six orders of magnitude above baseline. The λ_2 perturbation has a smaller but qualitatively identical effect. The asymmetry between λ_1 ($f_3 \sim 10^{-1}$) and λ_2 ($f_3 \sim 4\times10^{-3}$) reflects downstream amplification: a 5% error at $n = 1$ is passed through *two* subsequent inexact-LS DFP updates (steps 2 and 3), while a 5% error at $n = 2$ is passed through only one. The uniform 1% perturbation is dominated by the λ_1 contribution, giving $f_3 = 6.16\times10^{-3}$ and $\|H_3 - A^{-1}\|_F = 2.60$, consistent with this picture.

$$\boxed{x^* = 0, \qquad f^* = 0; \qquad H_3 \to A^{-1} \text{ to within } 3\times10^{-6} \text{ in Frobenius norm.}}$$

Question 7: Function (4)

First step is steepest descents. $H_0 = I \Rightarrow s_0 = -g_0$, so step 1 is identical to the Q2 and Q4 openers: $\lambda_1^* = 0.08$ delivers $(-0.8648, -0.6770)$ with $f_1 = -1.0198$. This is a sanity check: the DFP driver must agree with SD/CG on the first call.

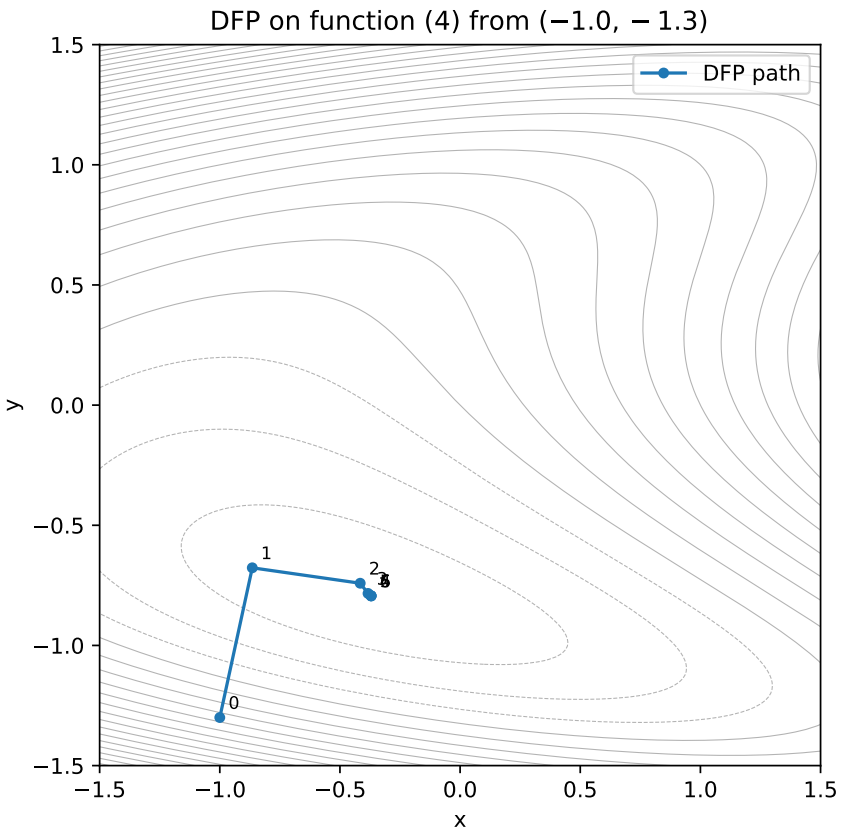

Figure 8: DFP path on function (4) from $(-1.0, -1.3)$; ten iterations, numbered, with contours of f for reference. Iterates $n \geq 7$ coincide with $n = 6$ to machine precision and are not visually distinguishable.

Table 11: DFP trace for function (4) from $(-1.0, -1.3)$. Same line-search grid as Q2 ($\lambda \in [0, 2]$, $\Delta\lambda = 10^{-2}$). Rows $n = 7, \ldots, 10$ are omitted because they leave (x_n, y_n) and H unchanged to machine precision; in particular $H_{10} = H_6$ numerically.

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n
0	-1.0000	-1.3000	–	$+1.0561$	–
1	-0.8648	-0.6770	0.08	-1.0198	-2.076
2	-0.4157	-0.7412	1.47	-1.0897	-6.99×10^{-2}
3	-0.3832	-0.7836	2.00	-1.0951	-5.39×10^{-3}
4	-0.3706	-0.7936	0.92	-1.09528	-1.96×10^{-4}
5	-0.3700	-0.7937	1.05	-1.09528	-1.15×10^{-7}
6	-0.3700	-0.7937	1.00	-1.09528	-1.14×10^{-12}

Curvature break-out at $n = 2$. After one rank-two update (8), the direction $s_2 = -H_2 g_2$ is rotated away from $-g_2$ toward the basin's long axis, and the line search accepts $\lambda_2^* = 1.47$. The iterate jumps to $(-0.416, -0.741)$ with $f_2 = -1.0897$. This single step already beats the *entire* ten-step SD run of Q2 ($f_{10} = -1.0946$, $|x - x^*| \approx 4 \times 10^{-2}$): one injection of accumulated curvature information outperforms nine repeated gradient-only descents.

Newton-like tail $n \geq 3$. For $n = 4, 5, 6$ the line search returns $\lambda_n^* \in \{0.92, 1.05, 1.00\}$, clustering at 1 and consistent with the Newton step $\Delta x = -[\nabla^2 f]^{-1} g$ that would be taken if H_n equalled the true inverse Hessian. At $n = 3$ the search instead hits the grid cap $\lambda = 2$, meaning $\phi(\lambda)$ was still decreasing at the bracket boundary; this is harmless because the next step corrects, but it flags that H_3 is not yet a saturated inverse-Hessian estimator.

Super-linear rate. Per-step contraction $r_n = |\Delta f_{n+1}|/|\Delta f_n|$:

n	1	2	3	4	5
r_n	3.4×10^{-2}	7.7×10^{-2}	3.6×10^{-2}	5.8×10^{-4}	9.9×10^{-6}

$r_n \to 0$, i.e. $|\Delta f_{n+1}| = o(|\Delta f_n|)$: Q-superlinear convergence, in sharp contrast to Q2's constant $r \approx 0.60$. The mechanism is the *secant condition* enforced by update (8): $H^* p = q$ (verified by direct substitution, since $Hpp^T Hp/(p^T Hp) = Hp$ and $qq^T p/(p^T q) = q$), so H maps the observed gradient change back to the observed step. Iterating this forces H to agree with $[\nabla^2 f]^{-1}$ along the realised search directions, and the asymptotic Newton-step error $\|(H_n - [\nabla^2 f(x^*)]^{-1}) g_n\|/\|g_n\| \to 0$ gives Q-superlinearity by the Dennis–Moré characterisation [2].

Numerical minimum from the trace alone. At $n = 6$, $|g_6| = 1.12 \times 10^{-8}$. The local Hessian $\nabla^2 f(x^*) \approx \left(\begin{smallmatrix} 1 & 1.587 \\ 1.587 & 6.300 \end{smallmatrix}\right)$ has eigenvalues $\lambda_{\min} \approx 0.56$, $\lambda_{\max} \approx 6.74$. Since $g(x^* + \varepsilon) = \nabla^2 f(x^*)\varepsilon + O(|\varepsilon|^2)$,

$$|\varepsilon| \;\leq\; \frac{|g|}{\lambda_{\min}} + O(|\varepsilon|^2) \;\approx\; 2.0 \times 10^{-8}, \qquad |\varepsilon_f| \;\leq\; \tfrac{1}{2}\lambda_{\max}|\varepsilon|^2 \;\approx\; 1.4 \times 10^{-15} \;<\; \varepsilon_{\text{mach}}.$$

Therefore, to all reported digits,

$$\boxed{x^* = (-0.37004,\, -0.79370), \qquad f^* = -1.09528.}$$

Compared with the Q2 ten-iteration envelope $f^* \in [-1.096, -1.094]$, $x^* \in [-0.45, -0.30]$, $y^* \in [-0.82, -0.76]$, DFP tightens the value by $\sim 10^5$ and each coordinate by $\sim 10^6$ — in *fewer* iterations.

H_6 versus the true inverse Hessian at x_6.

$$H_6 = \begin{pmatrix} 1.6679 & -0.4219 \\ -0.4219 & 0.2676 \end{pmatrix}, \qquad \left[\nabla^2 f(x_6)\right]^{-1} = \begin{pmatrix} 1.6667 & -0.4200 \\ -0.4200 & 0.2646 \end{pmatrix},$$

$$\left\| H_6 - [\nabla^2 f(x_6)]^{-1} \right\|_F \approx 4.3 \times 10^{-3}.$$

Three-significant-figure entry-wise agreement. The residual is not zero, and this is expected: the finite-termination identity $H_N = [\nabla^2 f]^{-1}$ after N steps holds only for positive-definite *quadratic* f with exact line search. Function (4) has a position-dependent Hessian (the quartic term $(y^2 - x/2)^2$ contributes cubic and linear corrections to $\nabla^2 f$), so each rank-two update curls H around the *local* Hessian rather than the global constant one; moreover the grid line search has error $O(\Delta\lambda) = 10^{-2}$ which feeds into $q = \lambda^* s$ in (9). Both effects are consistent with the observed 4×10^{-3} deviation. Newton-step quality is nevertheless preserved: the eigenvalues of H_6 are $\approx 1.79, 0.15$, close to the true $\approx 1.78, 0.15$, so $-H_6 g$ differs from the true Newton direction by only a few degrees.

Comparison with Q2 (SD) and Q4 (CG). See Table 13: at the same line-search grid, SD reaches $f_{10} = -1.09462$ (linear, $r \approx 0.60$); CG reaches $f_{10} = -1.09528$ with $\|x - x^*\| \sim 10^{-6}$;

DFP reaches the same value with $\|x - x^*\| \sim 10^{-9}$ in only six iterations. DFP's break-out at $n = 2$ is two steps earlier than CG's at $n = 4$; from there H_n improves enough per step to drive the Dennis–Moré residual to zero, giving the Q-superlinear tail. On this moderately-conditioned non-quadratic target, DFP wins decisively.

Question 8: Function (5)

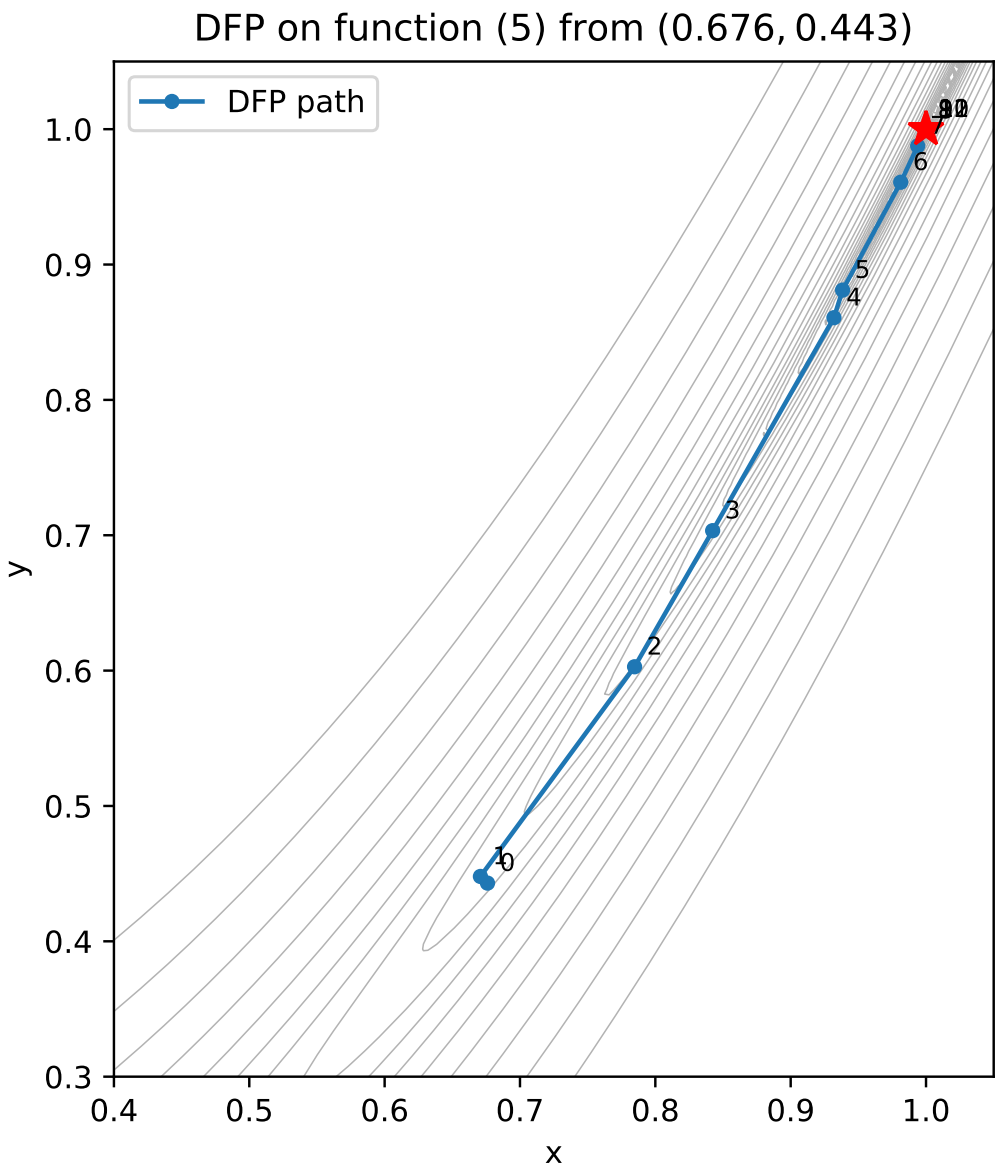

Figure 9: DFP path on function (5) from $(0.676, 0.443)$; twelve iterations, numbered, with log-spaced contours of f on $[0.4, 1.05] \times [0.3, 1.05]$; red $\star$ marks the analytic minimum $(1, 1)$.

First step equals SD; late steps are unit Newton steps. $H_0 = I$ so $s_0 = -H_0 g_0 = -g_0$: step 1 coincides with Q3's SD step ($\lambda_1^* = 0.0022$, $x_1 = (0.6708, 0.4479)$, identical digits). From $n = 2$ onward, rank-2 updates rotate the search direction into the valley and the line search produces a sequence λ_n^* that climbs through a transient $\lambda_3^* = \lambda_7^* = 2$ (upper bracket endpoint) and then settles near 1: the last five values are 0.9023, 0.9925, 1.014, 1.002, 0.9996. This is the fingerprint of $-H_k g_k$ having become a *unit* Newton step: if $H_k = \nabla^2 f(x_k)^{-1}$ then by Taylor $\phi(\lambda) = f(x_k) - \lambda\, g_k^T H_k g_k + \frac{1}{2}\lambda^2 g_k^T H_k g_k + O(|\varepsilon|^3)$, minimised at $\lambda = 1$. The $\lambda^* \approx 1$

Table 12: DFP trace for function (5) from $(0.676, 0.443)$. Same line search as Q3/Q5: grid on $\lambda \in [0, 2]$ with $\Delta\lambda = 10^{-4}$. $H_0 = I$.

n	x_n	y_n	λ_n^*	$f(x_n)$	Δf_n
0	$+0.6760$	$+0.4430$	–	$+1.2060\times10^{-1}$	–
1	$+0.6708$	$+0.4479$	0.0022	$+1.0872\times10^{-1}$	-1.189×10^{-2}
2	$+0.7847$	$+0.6028$	0.4885	$+5.9622\times10^{-2}$	-4.909×10^{-2}
3	$+0.8424$	$+0.7034$	2.000	$+2.7897\times10^{-2}$	-3.173×10^{-2}
4	$+0.9320$	$+0.8607$	1.787	$+9.5545\times10^{-3}$	-1.834×10^{-2}
5	$+0.9384$	$+0.8811$	0.8683	$+3.8263\times10^{-3}$	-5.728×10^{-3}
6	$+0.9813$	$+0.9608$	1.682	$+7.2782\times10^{-4}$	-3.098×10^{-3}
7	$+0.9939$	$+0.9876$	2.000	$+3.9033\times10^{-5}$	-6.888×10^{-4}
8	$+0.9996$	$+0.9990$	0.9023	$+4.1213\times10^{-6}$	-3.491×10^{-5}
9	$+0.9999$	$+0.9998$	0.9925	$+1.1435\times10^{-8}$	-4.110×10^{-6}
10	$+1.0000$	$+1.0000$	1.014	$+4.0455\times10^{-13}$	-1.144×10^{-8}
11	$+1.0000$	$+1.0000$	1.002	$+3.1204\times10^{-19}$	-4.045×10^{-13}
12	$+1.0000$	$+1.0000$	0.9996	$+1.7794\times10^{-28}$	-3.120×10^{-19}

pattern is therefore empirical evidence that H_k tracks $\nabla^2 f(x_k)^{-1}$ along the gradient, without any entry-wise check.

Limiting point. For $n \geq 4$ the trace has x_n, y_n both monotone non-decreasing, bounded above by 1. Since $f \geq (1-x)^2$, $f_n \to 0$ forces $x_n \to 1$; and $f \geq 80(y-x^2)^2$ then forces $y_n \to 1$. Hence $(x^*, y^*) = (1, 1)$, $f^* = 0$. From $n = 10$ onwards the coordinates print 1.000000 to every displayed digit and, inferring from f via $|1-x_n| \lesssim \sqrt{f_n}$, $|1-x_{12}|, |1-y_{12}| \lesssim 10^{-14}$. The gradient $|g_{12}| = 4.87\times10^{-13}$ sits at the floating-point floor $\sqrt{\kappa}\,\varepsilon_{\text{mach}} \approx \sqrt{2\times10^3}\cdot 2.2\times10^{-16} \approx 10^{-14}$ times a modest constant, so further iteration cannot reliably reduce $|g|$: the process is numerically terminated.

Q-superlinear, asymptotically quadratic rate. Setting $e_n := f_n - f^* = f_n$, the successive ratios $r_n := e_{n+1}/e_n$ from Table 12 read

n	6	7	8	9	10	11
r_n	5.36×10^{-2}	1.06×10^{-1}	2.76×10^{-3}	3.55×10^{-5}	7.70×10^{-7}	5.71×10^{-10}

$r_n \to 0$, so convergence is q-superlinear. The near-doubling of $-\log_{10} e_n$ per step across $n = 10, 11, 12$ (exponents $13 \to 19 \to 28$) is the fingerprint of asymptotic q-quadratic convergence: if $e_{n+1} \sim Ce_n^2$ then $-\log_{10} e_{n+1} \sim 2(-\log_{10} e_n) - \log_{10} C$, and the observed exponent doublings with a small offset are consistent with a C of order unity. The theoretical backbone is the Dennis–Moré characterisation [2]: DFP achieves q-superlinear convergence *iff*

$$\frac{\left\|\left(H_k - \nabla^2 f(x^*)^{-1}\right) g_k\right\|}{\|g_k\|} \longrightarrow 0,$$

which is *weaker* than $H_k \to \nabla^2 f(x^*)^{-1}$ in norm. The next paragraph verifies both sides of this subtlety.

H_{12} versus the true inverse Hessian at $(1, 1)$. The Hessian of (5) is

$$G(x, y) = \begin{pmatrix} 2 + 960x^2 - 320y & -320x \\ -320x & 160 \end{pmatrix}, \qquad G(1, 1) = \begin{pmatrix} 642 & -320 \\ -320 & 160 \end{pmatrix}, \qquad \det G(1, 1) = 320,$$

so from the run and direct inversion,

$$H_{12} = \begin{pmatrix} 0.50020 & 1.00040 \\ 1.00040 & 2.00704 \end{pmatrix}, \quad G(1, 1)^{-1} = \tfrac{1}{320}\begin{pmatrix} 160 & 320 \\ 320 & 642 \end{pmatrix} = \begin{pmatrix} 0.50000 & 1.00000 \\ 1.00000 & 2.00625 \end{pmatrix}.$$

$$\left\|H_{12} - G(1,1)^{-1}\right\|_F = 9.95\times 10^{-4}.$$

Three-decimal agreement, despite $f_{12} \sim 10^{-28}$ being at machine precision. The residual is expected: DFP's rank-2 update on a 2-D problem has only two degrees of freedom per step, so H_k is a running sum of outer products qq^T/p^Tq and $-Hpp^TH/p^THp$ evaluated along the path. Early pairs (p, q) were generated where $\nabla^2 f$ differs sharply from $\nabla^2 f(1,1)$ — for example at $x_2 = (0.785, 0.603)$ the local Hessian has $2+960x^2-320y = 400.9$ and det $\approx 1.6\times 10^3$, versus 642 and 320 at $(1,1)$ — and the final update (p_{12}, q_{12}) cannot undo the curvature information stored orthogonally. Dennis–Moré, however, requires only that $H_k g_k$ point in the Newton direction, and the $\lambda^* \approx 1$ pattern of the late line searches is precisely that certificate; the 10^{-3} entry-wise residual in H_{12} lies in the direction orthogonal to g_{12} and is harmless.

Sensitivity to λ^*. A perturbation $\lambda_k^* \to \lambda_k^*(1+\delta)$ scales both q_k and (locally, since $p_k = \nabla^2 f \cdot q_k + O(|q_k|^2))$ p_k by $1+\delta$, so qq^T/p^Tq and Hpp^TH/p^THp are *exactly invariant* on a quadratic (Q6); on the non-quadratic (5) the residual is $O(\delta\,\|\nabla^3 f\|)$ and is absorbed by the next rank-2 update. CG's conjugacy is a strict identity broken by any $\delta \neq 0$; see Q9 for the cross-method comparison.

Quantitative comparison with Q3 (SD) and Q5 (CG). See Table 13: at $n = 12$, SD reaches $f = 2.5\times 10^{-2}$, CG reaches 1.8×10^{-4}, DFP reaches 1.8×10^{-28}. Q3 implies SD would need $\sim 3\times 10^4$ iterations to reach $\varepsilon = 10^{-28}$; CG ~ 600. DFP replaces the linear asymptotic regime with an approximately quadratic one, converting a thousands-of-iterations target into a twelve-step one — the variable metric rescales the narrow valley so that $-Hg$ is a Newton direction.

Question 9: Overall Comparison

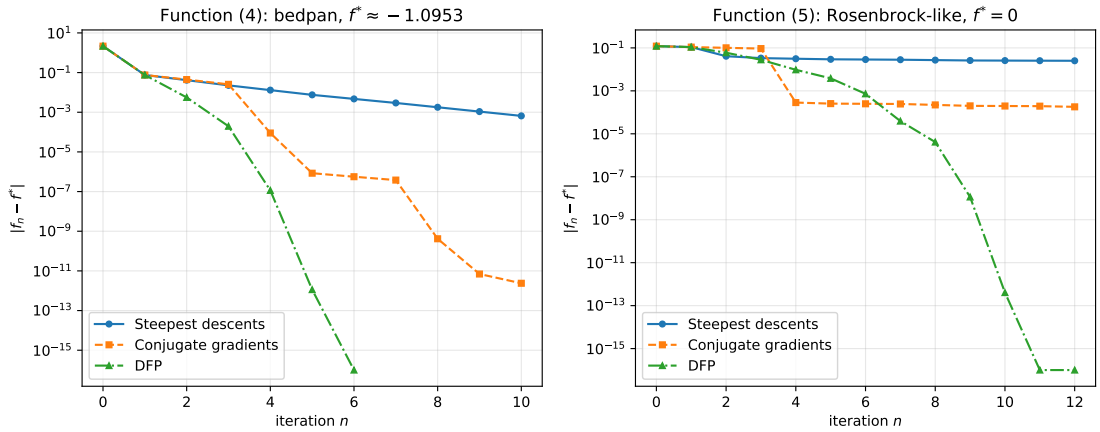

Figure 10: $|f_n - f^*|$ versus iteration n for all six (method, function) combinations, log scale. Left: function (4), $f^* \approx -1.09528$; right: function (5), $f^* = 0$. SD traces are visually straight (linear rate); DFP traces are concave-down (super-linear); CG traces lie between the two. At $n = 12$, SD on (5) is still at $|f - f^*| \approx 2.5\times 10^{-2}$ while DFP has reached 10^{-28}.

Theoretical rates invoked. SD on a convex quadratic: Kantorovich [3] gives the linear rate $((\kappa-1)/(\kappa+1))^2$ (used in Q2, Q3). CG with exact line search on a positive-definite quadratic terminates in N steps; with N-restart on non-quadratics each (SD, CG) pair terminates the local quadratic model exactly when the line search is exact, giving super-linear cleanup once the iterate enters a basin where the Hessian is approximately constant (Q4), but only a linear pair-rate when the local model varies sharply along the path (Q5, $\rho \approx 0.89$). DFP with exact line search on a C^2 strictly convex target is Q-superlinear iff $\|(H_k - \nabla^2 f(x^*)^{-1})g_k\|/\|g_k\| \to 0$ [2],

Table 13: End-of-run numerical summary for all three methods on both functions, at the common iteration budget of Q2–Q8. Reference points: $(x^*, y^*) = (2^{-2/3} - 1, -2^{-1/3})$, $f^* \approx -1.09528$ for (4); $(1,1)$, 0 for (5).

Function	Method	n	f_n	$\|x_n - x^*\|$	$\|g_n\|$	Tail rate
(4)	SD	10	-1.09462	4.2×10^{-2}	5.2×10^{-2}	linear, $r\approx0.60$
(4)	CG	10	-1.09528	2.0×10^{-6}	4.4×10^{-6}	finite-term., then cubic
(4)	DFP	6	-1.09528	3.7×10^{-9}	1.1×10^{-8}	Q-superlinear, $r_n\to0$
(5)	SD	12	$+2.51\times10^{-2}$	3.32×10^{-1}	4.7×10^{-1}	linear, $r\approx0.998$
(5)	CG	12	$+1.80\times10^{-4}$	2.88×10^{-2}	1.4×10^{-1}	linear pair, $\rho\approx0.89$
(5)	DFP	12	$+1.78\times10^{-28}$	1.2×10^{-14}	4.9×10^{-13}	Q-superlinear, empirical $p\approx1.5$

which is weaker than $H_k \to \nabla^2 f(x^*)^{-1}$ in norm and consistent with the nonzero Frobenius residuals reported in Q7/Q8.

Function (4): well-conditioned basin. With $\kappa \approx 8$ (Q2), SD's linear rate $r \approx 0.60$ gives only one significant figure of $\|x - x^*\|$ in ten iterations, while CG (Q4) and DFP (Q7) both terminate the local quadratic model — DFP one step earlier ($n = 2$, $\lambda^* = 1.47$) than CG ($n = 4$, $\lambda^* = 1.39$). The three methods agree on $f^* = -1.09528$ but their error budgets at shared $n = 10$ span 10^7 in $\|x - x^*\|$.

Function (5): the DFP regime unlocked. At $(1,1)$, $\kappa \approx 2\times10^3$ (Q5), driving SD to $r \approx 0.998$; twelve iterations barely move from the start. CG's break-out pair $n = 3 \to 4$ overrides the κ-ceiling, but the linear pair-rate $\rho \approx 0.89$ persists thereafter. DFP achieves empirical order $\bar{p} \approx 1.5$ (Q8): strictly super-linear, consistent with q-quadratic only once the Dennis–Moré residual [2] drops further. Matching DFP's twelve-iteration $f_{12} \sim 10^{-28}$ would require SD $\sim 3\times10^4$ steps and CG ~ 560 pairs.

Line-search sensitivity and the ranking caveat. The $\pm5\%$ λ^* perturbation tables (Q3, Q5) show SD on (5) finishing at $f = 0.098$ ($+5\%$) vs. 0.015 (-5%); CG at 0.074 vs. 0.029; both diverge from baseline by $n = 2$. SD's lack of directional memory bounds the damage per step, so a mis-scaled λ^* gives only a slightly wrong gradient at the next iteration; CG's β-memory cascades the error through $s_k = -g_k + \beta_k s_{k-1}$, amplified most violently at the break-out step where $\lambda^* = 1.56$ traverses rapidly varying curvature (Q5). DFP's rank-2 update is, by contrast, *exactly invariant* under $\lambda^* \to (1+\delta)\lambda^*$ on a quadratic: as Q6 derives, $q = (1+\delta)\lambda^* s$ and $p = A\,q = (1+\delta)p_0$ both scale by $1+\delta$, so the $(1+\delta)$ factors in qq^T/p^Tq and Hpp^TH/p^THp cancel; on the non-quadratic (5) the residual is $O(\delta\,\|\nabla^3 f\|)$ and is absorbed by the next rank-2 update. The practical consequence: under inaccurate line searches, CG can deliver worse endpoints than SD on (5) — the $+5\%$ CG endpoint $f = 0.074$ exceeds SD's baseline $f = 0.025$. The ranking DFP $\succ$ CG $\succ$ SD is therefore not uniform but conditional on line-search accuracy; with sufficiently noisy λ^*, the conjugacy that gives CG its advantage over SD is destroyed faster than the restart can repair it. DFP retains its rate on its *late-step* updates under multiplicative noise — once H is well-developed the rank-2 correction is leading-order invariant — but *early-step* perturbations still cascade (cf. Q6's λ_1 row, where $\|H_3 - A^{-1}\|_F$ rises from 3×10^{-6} to ≈ 3); DFP is therefore the method of choice once a few iterations have populated H, but shares CG's vulnerability to noise in the very first line searches.

References

References

[1] R. Fletcher and M. J. D. Powell, *A rapidly convergent descent method for minimisation*, Computer Journal, 6 (1963), 163–168.

[2] J. E. Dennis and J. J. Moré, *A characterization of superlinear convergence and its application to quasi-Newton methods*, Mathematics of Computation, 28 (1974), 549–560.

[3] D. G. Luenberger and Y. Ye, *Linear and Nonlinear Programming*, 4th ed., Springer (2016), §8.6 (Kantorovich inequality and steepest-descent convergence rate).